Part 4 file handoff agreement

Scope. The tool and model team exchange UTF-8 JSON, JSONL and CSV files using the same source images; packages do not copy images. No HTTP API, real training or weights execution is included. Optional `weights_ref` is an opaque provenance string. Single-page PDF; reviewer runbook.

Authoritative contracts. Annotation V1, training/review package, and model return define exact types, required fields and forbidden extra fields. The runbook specifies publishing and legacy compatibility.

Tool to model team. `training_update_v2` (package schema 2) uses `verified_only`: only frames verified against current content, including unchanged and negative frames. Empty coverage or an unknown baseline is rejected. `review_export_v1` (package schema 1) uses `all_frames_review` and preserves actual review status. Both keep annotation schema V1 and exactly six files: `manifest.json`, `data/corrected_annotations.jsonl`, `data/frame_map.jsonl`, `reports/diff.json`, `reports/diff.csv`, `reports/summary_by_class.csv`.

Manifest and identity. Read full package ID, round/model/taxonomy, media/source identity, complete source frame map, baseline digest, coverage, verification and revision from the manifest. Artifact entries declare relative path, bytes and SHA-256. IDs exclude revision, destination and creation time. New manifests include UTC-second `created_at`; validated legacy packages may omit it. 接收方须先升级两端校验器；旧校验器会拒绝该字段。 A baseline digest identifies provenance; it cannot authenticate unseen original predictions.

Annotation records. Required: `schema_version:1`, nonempty `source`, nonnegative integer `frame`, and `regions` (empty means no regions). Optional nonnegative `time_s` stays absent when omitted; if supplied it must exactly match Source. Use original frame IDs, not playback indices. Each region requires nonempty `id`, `class`, `kind`; at least one geometry: pixel `box:[x,y,width,height]` (nonnegative origin, positive size) or `polygon:[[x,y],...]` (at least three nonnegative 2D vertices). Both geometries may coexist. Optional `conf` is 0-1; `track_id` accepts a string (including empty) or null. Corrected exports use `source:human_corrected`; model returns use `manifest.media.source`. Sample IDs are `<media_id>_<frame_id:06d>`.

Model team to tool. Return `manifest.json` plus its referenced `model_output_v1.jsonl`. Manifest requires `schema_version:1`, `package_type:model_round_v1`, `annotation_schema_version:1`, a new `round_id`, `model_revision`, `parent_package_id`, unchanged `taxonomy_version`, `media`, complete `source_frame_entries`, and `annotations:{path,bytes,sha256}` with fixed path `model_output_v1.jsonl`. Supply every Source frame exactly once, including frames excluded from training. Weights alone are insufficient. Supply the actual parent training package directory; its integrity, content, old round, baseline, taxonomy and full frame map must match.

Acceptance and transition. Reject malformed UTF-8 before parsing, unknown fields, mismatched bytes/digests/identity/coverage, or same-round returns. Preview preserves active state. Commit revalidates inputs, archives exact saved V3 bytes as `label/rounds/<sha256>.json`, then atomically replaces active V3. Failure preserves active data. New sessions start at revision zero with empty verification, batch and undo history. No schema-version bump is implied by a new model round.